

Syllabus

UNIT I : ARM7, ARM9, ARM11 Processors

Introduction to ARM processors and its versions, ARM7, ARM9 & ARM11 features, advantages & suitability in embedded application, ARM7 data flow model, programmer's model, modes of operations, Instruction set, programming in assembly language.

(Refer chapter 1)

UNIT II : ARM7 Based Microcontroller

ARM7 Based Microcontroller LPC2148: Features, Architecture (Block Diagram and Its Description), System Control Block (PLL and VPB divider) , Memory Map, GPIO, Pin Connect Block, timer, interfacing with LED, LCD, GLCD, KEYPAD. (Refer chapter 2)

UNIT III : Real World Interfacing with ARM7 Based Microcontroller

Interfacing the peripherals to LPC2148: GSM and GPS using UART, on-chip ADC using interrupt (VIC), EEPROM using I2C, SDCARD using SPI, on-chip DAC for waveform generation.

(Refer chapter 3)

UNIT IV : ARM CORTEX Processors

Introduction to ARM CORTEX series, improvement over classical series and advantages for embedded system design. CORTEX A, CORTEX M, CORTEX R processors series, versions, features and applications. Need of operating system in developing complex applications in embedded system, desired features of operating system & hardware support from processor, Firmware development using CMSIS standard for ARM Cortex. Survey of CORTEX M3 based controllers, its features and comparison. ✓

(Refer chapter 4)

UNIT V : ARM CORTEX M3 based Microcontroller

ARM-CM3 Based Microcontroller LPC1768: Features, Architecture (Block Diagram & Its Description), System Control, Clock & Power Control, GPIO, Pin Connect Block, interfacing with RGB LED, Seven Segment, TFT Display, MOTOR control using PWM. (Refer chapter 5)

UNIT VI : Real World Interfacing with ARM-CM3 Based Microcontroller

Concept of USB, CAN, and Ethernet based communication using microcontrollers. CAN, USB, ETHERNET applications in embedded c.

(Refer chapter 6)



UNIT - II

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ARM7 Based Microcontroller

2.1 LPC 2148

† The LPC2141/42/44/46/48 microcontrollers are based on a 16-bit/32-bit ARM7TDMI-S CPU with real-time emulation and embedded trace support, that combine the microcontroller with embedded high-speed flash memory ranging from 32 KB to 512 KB. A 128-bit wide memory interface and a unique accelerator architecture enable 32-bit code execution at the maximum clock rate. For critical code size applications, the alternative 16-bit thumb mode reduces code by more than 30 % with minimal performance penalty. Due to their tiny size and low power consumption, LPC2141/42/44/46/48 are ideal for applications where miniaturization is a key requirement, such as access control and point-of-sale. Serial communications interfaces ranging from a USB 2.0 Full-speed device, multiple UARTs, SPI, SSP to I²C-bus and on-chip SRAM of 8 KB up to 40 KB, make these devices very well suited for communication gateways and protocol converters, soft modems, voice recognition and low end imaging, providing both large buffer size and high processing power. Various 32-bit timers, single or dual 10-bit ADC(s), 10-bit DAC, PWM channels and 45 fast GPIO lines with up to nine edge or level sensitive external interrupt pins make these microcontrollers suitable for industrial control and medical systems.

2.2 † Features of LPC 2148

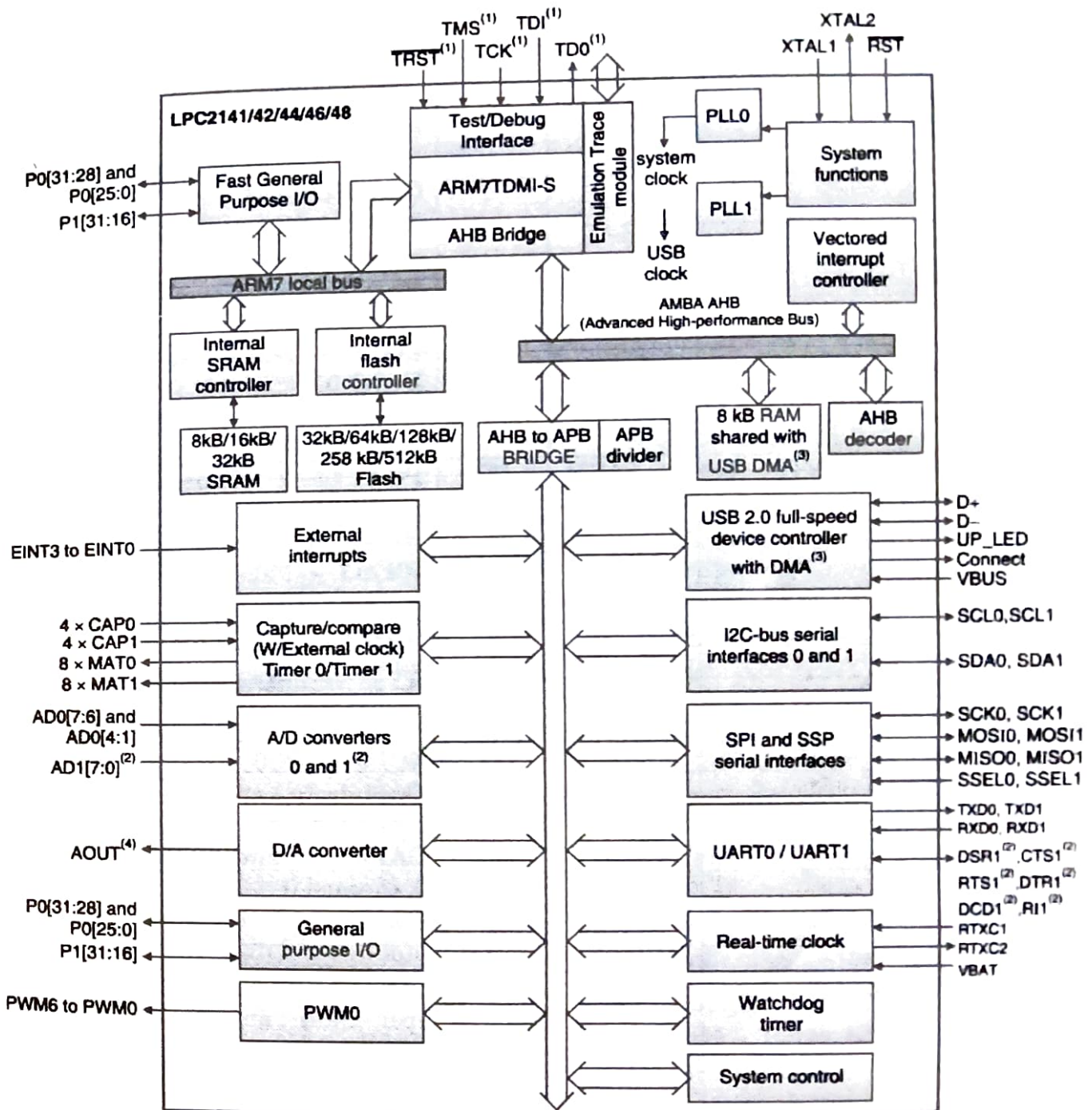
- 8 KB to 40 KB of on-chip static RAM and 32 KB to 512 KB of on-chip flash memory. 128-bit wide interface/accelerator enables high-speed 60 MHz operation.



- In-System Programming/In-Application Programming (ISP/IAP) via on-chip boot loader software. Single flash sector or full chip erase in 400 ms and programming of 256 B in 1 ms.
- Embedded ICE RT and Embedded Trace interfaces offer real-time debugging with the on-chip Real Monitor software and high-speed tracing of instruction execution.
- USB 2.0 Full-speed compliant device controller with 2 kB of endpoint RAM. In addition, the LPC2146/48 provides 8 KB. of on-chip RAM accessible to USB by DMA.
- One or two (LPC2141/42 vs. LPC2144/46/48) 10-bit ADCs provide a total of 6/14 analog inputs, with conversion times as low as 2.44 μ s per channel.
- Single 10-bit DAC provides variable analog output (LPC2142/44/46/48 only).
- Two 32-bit timers/external event counters (with four capture and four compare channels each), PWM unit (six outputs) and watchdog.
- Low power Real-Time Clock (RTC) with independent power and 32 kHz clock input.
- Multiple serial interfaces including two UARTs (16C550), two Fast I²C-bus (400 kbit/s), SPI and SSP with buffering and variable data length capabilities.
- Vectored Interrupt Controller (VIC) with configurable priorities and vector addresses.
- Up to 45 of 5 V tolerant fast general purpose I/O pins in a tiny LQFP64 package.
- Up to 21 external interrupt pins available.
- 60 MHz maximum CPU clock available from programmable on-chip PLL with settling time of 100 μ s.
- On-chip integrated oscillator operates with an external crystal from 1 MHz to 25 MHz.
- Power saving modes include Idle and Power-down.
- Individual enable/disable of peripheral functions as well as peripheral clock scaling for additional power optimization.
- Processor wake-up from Power-down mode via external interrupt or BOD.
- Single power supply chip with POR and BOD circuits:
 - CPU operating voltage range of 3.0 V to 3.6 V ($3.3 \text{ V} \pm 10 \%$) with 5 V tolerant I/O pads.

2.3 Architecture (Block Diagram and its Description)

PU - Oct. 2011, April 2012, Oct. 2012, May 2015



† Fig. 2.3.1 : Block diagram

† Courtesy NXP/Philips



† Table 2.3.1 : Pin description (Courtesy NXP)

Symbol	Pin	Type	Description
P0.0 to P0.31		I/O	Port 0 : Port 0 is a 32-bit I/O port with individual direction controls for each bit. Total of 31 pins of the Port 0 can be used as a general purpose bidirectional digital I/Os while P0. 31 is output only pin. The operation of port 0 pins depends upon the pin function selected via the pin connect block. Pins P0.24, P0.26 and P0.27 are not available.
P0.0/TXD0/PWM1	19	I/O	P0.0 - General purpose input/output digital pin (GPIO).
		O	IXD0 - Transmitter output for UART 0.
		O	PWM1 - Pulse With Modulator output 1.
P0.1/RXD0/PWM3/EINT0	21	I/O	P0.1 - General purpose input/output digital pin (GPIO).
		I	RXD0 - Receiver input for UART 0
		O	PWM3 - Pulse Width Modulator output 3.
		I	EINT0 - External interrupt 0 input.
P0.2/SCL0/CAP0.0	22	I/O	P0.2 - General purpose input/output digital pin (GPIO).
		I/O	SCL0 - I ² C0 clock input/output. Open-drain output (for I ² C-bus compliance)
		I	CAP0.0 - Capture input for Timer 0, channel 0
P0.3/SDA0/ MAT0.0/EINT1	26	I/O	P0.3 - General purpose input/output digital pin (GPIO).
		I/O	SDA0 - I ² C0 data input/output open-drain output (for I ² C-bus compliance).
		O	MAT0.0 - Match output for Timer 0, channel 0
		I	EINT1 - External interrupt 1 input

Symbol	Pin	Type	Description
P0.4/SCK0/CAP0.1/AD0.6	27	I/O	P0.4 – General purpose input/output digital pin (GPIO)
		I/O	SCK0 – Serial clock for SPI0. SPI clock output from master or input to slave.
		I	CAP0.1 – Capture input for Timer 0. Channel 1
		I	AD0.6 – ADC0, input 6
P0.5/MISO0/MAT0.1/AD0.7	29	I/O	P0.5 – General purpose input/output digital pin (GPIO)
		I/O	MISO0 – Master in Slave Out for SPI0. Data input to SPI master or data output from SPI slave
		O	MAT0.1 – Match output for Timer 0, channel 1
		I	AD0.7 – ADC0, input 7
P0.6/MOSI0/CAP0.2/AD1.0	30	I/O	P0.6 – General purpose input/output digital pin (GPIO)
		I/O	MOSI0 – Master out slave in for SPI0. Data output from SPI master or data input to SPI slave.
		I	CAP0.2 – Capture input for Timer 0, channel 2
		I	AD1.0 – ADC1, input 0. Available in LPC2144/46/48 only.
P0.7/SSEL0/PWM2/EINT2	31	I/O	P0.7 – General purpose input/output digital pin (GPIO)
		I	SSEL0 – Slave select for SPI0. Selects the SPI interface as a slave.
		O	PWM2 – Pulse width modulator output 2
		I	EINT2 – External interrupt 2 input
P0.0/TXD1/PWM4/AD1.1	33	I/O	P0.0 – General purpose input/output digital pin (GPIO)
		O	TXD1 Transmitter output for UART1.
		O	PWM4 – Pulse width modulator output 4
		I	AD1.1 – ADC1, input 1. Available in LPC2144/46/48 only.



Symbol	Pin	Type	Description
P0.9/RXD1/PWM6/EINT3	34	I/O	P0.9 - General purpose input/output digital pin (GPIO)
		I	RXD1 Receiver input for UART1
		O	PWM6 - Pulse width modulator output 6
		I	EINT3 - External interrupt 3 input
P0.10/RTS1/CAP1.0/AD1.2	35	I/O	P0.10 - General purpose input/output digital pin (GPIO)
		O	RTS1 - Request to send output for UART1. LPC2144/46/48 only.
		I	CAP1.0 - capture input for Timer 1, channel 0
		I	AD1.2 - ADC 1, input 2. available in LPC2144/46/48 only.
P0.11/CTS1/CAP1.1/SCL1	37	I/O	P0.11 - General purpose input/output digital pin (GPIO)
		I	CTS1 - Clear to send input for UART1. Available in LPC2144/46/48 only.
		I	CAP1.1 - Capture input for Timer 1, Channel 1
		I/O	SCL1 - I ² C1 clock input/output. Open drain output (for I ² C bus compliance)
P0.12/DSR1/MAT1.0/AD1.3	38	I/O	P0.12 - General purpose input/output digital pin (GPIO).
		I	DSR1 - Data set Ready input for UART1. Available in LPC2144/46/48 only.
		O	MAT1.0 - Match output for Timer 1, channel 0.
		I	AD1.3 - ADC 1 input 3. Available in LPC2144/46/48 only.
P0.13/DTR1/MAT1.1/AD1.4	39	I/O	P0.13 - General purpose input/output digital pin (GPIO)
		O	DTR1 - Data Terminal ready output for UART1. LPC2144/46/48 only.
		O	MAT1.1 - Match output for Timer 1, channel 1.
		I	AD1.4 - ADC 1 input 4. Available in LPC2144/46/48 only.

Symbol	Pin	Type	Description
P0.14/DCD1/EINT1/SDA1	41	I/O	P0.14 - General purpose input/output digital pin (GPIO)
		I	DCD1 - Data carrier detect input for UART1. LPC2144/46/48 only.
		I	EINT1 - External interrupt 1 input
		I/O	SDA1 - I ² C1 data input/output. Open-drain output (for I ² C-bus compliance) Note : LOW on this pin while RESET is LOW forces on-chip boot loader to take over control of the part after reset.
P0.15/RI1/EIN12/AD1.5	45	I/O	P0.15 - General purpose input/output digital pin (GPIO)
		I	RI1 - Ring indicator input for UART1. Available in LPC2144/46/48 only.
		I	EINT2 - External interrupt 2 input
		I	AD1.5 - ADC1 input 5 Available in LPC2144/46/48 only.
P0.16/EINT0/MAT0.2/CAP0.2	46	I/O	P0.16 - General purpose input/output digital pin (GPIO).
		I	EINT0 - External interrupt 0 input
		O	MAT0.2 - Match output for Timer 0, Channel 2
		I	CAP0.2 - Capture input or Timer 0, Channel 2.
P0.17/CAP1.2/SCK1/MAT1.2	47	I/O	P0.17 - General purpose input/output digital pin (GPIO).
		I	CAP1.2 - Capture input for Timer 1, channel 2
		I/O	SCK 1 - Serial clock for SSP. Clock output from master or input to slave.
		O	MAT1.2 - Match output for Timer 1. Channel 2.
P0.18/CAP1.3/MISO1/MAT1.3	53	I/O	P0.18 - General purpose input/output digital pin (GPIO)
		I	CAP1.3 - Capture input for Timer 1, Channel 3



Symbol	Pin	Type	Description
		I/O	MISO1 - Master In slave out for SSP. Data input to SPI master or data output from SSP slave.
		O	MAT1.3 - Match output for Timer 1, channel 3.
P0.19/MAT1.2/MOSI1/CAP1.2	54	I/O	P0.19 - General purpose input/output digital pin (GPIO)
		O	MAT1.2 - Match output for Timer 1, channel 2
		I/O	MOSI1 - Master out slave in for SSP. Data output from SSP master or data input to SSP slave.
		I	CAP1.2 - Capture input for Timer 1, channel 2
P0.20/MAT1.3/SSEL1/EINT3	55	I/O	P0.20 - General purpose input/output digital pin (GPIO)
		O	MAT1.3 - Match output for Timer 1, channel 3.
		I	SSEL1 - Slave select for SSP. Selects the SSP interface as a slave.
		I	EINT3 - External interrupt 3 input.
P0.21/PWM5/AD1.6/CAP1.3	1	I/O	P0.21 - General purpose input/output digital pin (GPIO)
		O	PWM5 - Pulse width Modulator output 5
		I	AD1.6 - ADC 1, input 6 Available in LPC2144/46/48 only
		I	CAP1.3 - Capture input for Timer 1, channel 3
P0.22/AD1.7/CAP0.0/MAT0.0	2	I/O	P0.22 - General purpose input/output digital pin (GPIO)
		I	AD1.7 - ADC 1, input 7. Available in LPC2144/46/48 only.
		I	CAP0.0 - Capture input for Timer 0, Channel 0
		O	MAT0.0 - Match output for Timer 0, Channel 0.

Symbol	Pin	Type	Description
P0.23/V _{BUS}	58	I/O	P0.23 - General purpose input/output digital pin (GPIO)
		I	V_{BUS} - Indicates the presence of USB bus power Note : This signal must be HIGH for USB reset to occur.
P0.25/AD0.4/AOUT	9	I/O	P0.25 - General purpose input/output digital pin (GPIO)
		I	AD0.4 - ADC 0, input 4
		O	AOUT - DAC output. Available in LPC2142/44/46/48 only
P0.28/AD0.1/CAP0.2/MAT0.2	13	I/O	P0.28 - General purpose input/output digital pin (GPIO)
		I	AD0.1 - ADC 0, input 1
		I	CAP0.2 - Capture input for Timer 0, Channel 2
		O	MAT0.2 - match output for Timer 0, channel 2
P0.29/AD0.2/CAP0.3/MAT0.3	14	I/O	P0.29 - General purpose input/output digital pin (GPIO)
		I	AD0.2 - ADC.0 input 2
		I	CAP0.3 - Capture input for Timer 0, Channel 3
		O	MAT0.3 - Match output for Timer 0, Channel 3.
P0.30/AD0.3/EINT3/CAP0.0	15	I/O	P0.30 - General purpose input/output digital pin (GPIO)
		I	AD0.3 - ADC 0, input 3
		I	EINT3 - External interrupt 3 input
		I	CAP0.0 - Capture input for Timer 0, channel 0
P0.31/UP_LED/CONNECT	17	O	P0.31 - General purpose output only digital pin (GPO)
		O	UP_LED - USB Goodlink LED indicator. it is LOW when device is configured (non-control endpoints enabled). It is HIGH when the device is not configured or during global suspend.



Symbol	Pin	Type	Description
		O	CONNECT – Signal used to switch an external 1.5 kΩ resistor under the software control. Used with the softconnect USB feature. Important : This is an digital output only pin. This. pin MUST NOT be externally pulled LOW when RESET pin is LOW or the JTAG port be disabled.
P1.0 to P1.31		I/O	Port 1 : Port 1 is a 32-bit bidirectional I/O port with individual direction controls for each bit. The operation of port 1 pins depends upon the function selected via the pin connected block. Pins 0 through 15 of port 1 are not available.
P1.16/TRACEPKT0	16	I/O	P1.16 – General purpose input/output digital pin (GPIO). Standard I/O port with internal pull-up.
		O	TRACEPKT0 – Trace Packet, bit 0
P1.17/TRACEPKT1	12	I/O	P1.17 – General purpose input/output digital pin (GPIO) standard I/O port with internal pull up.
		O	TRACEPKT1 – Trace packet. bit 1
P1.18/TRACEPKT2	8	I/O	P1.18 – General purpose input/output pin (GPIO) standard I/O port with internal pull-up
		O	TRACEPKT2 – Trace packet, bit 2
P1.19/TRACEPKT3	4	I/O	P1.19 – General purpose input/output digital pin (GPIO) standard I/O port with internal pull up.
		O	TRACEPKT3 – Trace packet. bit 3
P1.20/TRACESYNC	48	I/O	P1.20 – General purpose input/output digital pin (GPIO). Standard I/O port with internal pull-up
		O	TRACESYNC – Trace synchronization

Symbol	Pin	Type	Description
			Note : LOW on this pin while $\overline{\text{RESET}}$ is LOW enables pin P1.25 : 16 to operate as Trace port after reset.
P1.21/PIPESTAT0	44	I/O	P1.21 - General purpose input/output digital pin (GPIO). Standard I/O port with internal pull-up.
		O	PIPESTAT0 - Pipeline status, bit 0
P1.22/PIPESTAT1	40	I/O	P1.22 - General purpose input/output digital pin (GPIO) standard I/O port with internal pull-up
		O	PIPESIAI0 - Pipeline status, bit 0
P1.23/PIPESTAT2	36	I/O	P1.23 - General purpose input/output digital pin (GPIO), standard I/O port with internal pull-up
		O	PIPESTAT2 - Pipeline status, bit 2
P1.24/TRACECLK	32	I/O	P1.24 - General purpose input/output digital pin (GPIO) standard I/O port with internal pull-up
		O	TRACECLK - Trace clock.
P1.25/EXTIN0	28	I/O	P1.25 - General purpose input/output digital pin (GPIO). Standard I/O port with internal pull-up
		I	EXTIN0 - External Trigger input



Symbol	Pin	Type	Description
P1.26/RTCK	24	I/O	P1.26 - General purpose input/output digital pin (GPIO)
		I/O	RTCK - Returned test clock output. Extra signal added to the JTAG port. Assists debugger synchronization when processor frequency varies Bidirectional pin with internal pull-up. Note : LOW on RTCK while $\overline{\text{RESET}}$ is LOW enables pin P1[31:26] to operate as Debug port after reset.
P1.27/TDO	64	I/O	P1.27 - General purpose input/output digital pin (GPIO)
		O	TDO - Test Data out for JTAG interface
P1.28/TDI	60	I/O	P1.28 - General purpose input/output digital pin (GPIO)
		I	TDI - Test data in for JTAG interface
P1.29/TCK	56	I/O	P1.29 general purpose input/output digital pin (GPIO).
		I	TCK - Test Clock for JTAG interface. This clock must be slower than $\frac{1}{8}$ of the CPU clock (CCLK) for the JTAG interface to operate.
P1.30/TMS	52	I/O	P1.30 - General purpose input/output digital pin (GPIO).
		I	TMS - Test Mode Select for JTAG interface.
P1.31/ $\overline{\text{TRST}}$	20	I/O	P1.31 - General purpose input/output digital pin (GPIO).
			$\overline{\text{TRST}}$ - Test Reset for JTAG interface.
D+	10	I/O	USB bidirectional D+ line.
D-	11	I/O	USB bidirectional D+ line.
$\overline{\text{RESET}}$	57	I	External reset input : A LOW on this pin resets the device, causing I/O ports and peripherals to take on their default states, and processor execution to begin at address 0. TTL with hysteresis, 5 V tolerant.

Symbol	Pin	Type	Description
XTAL1	62	I	Input to the oscillator circuit and internal clock generator circuits.
XTAL2	61	O	Output from the oscillator amplifier.
RTCX1	3	O	Input to the RTC oscillator circuit.
RTCX2	5	O	Output from the RTC oscillator circuit.
V _{SS}	6, 18, 25, 42, 50	I	Ground : 0 V reference
V _{SSA}	59	I	Analog ground : 0 V reference. This should nominally be the same voltage as V _{SS} , but should be isolated to minimize noise and error.
V _{DD}	23, 43, 51	I	3.3 V power supply : This is the power supply voltage for the core and I/O ports.
V _{DDA}	7	I	Analog 3.3 V power supply : This should be nominally the same voltage as V _{DD} but should be isolated to minimize noise and error. This voltage is only used to power the on-chip ADC(s) and DAC.
V _{REF}	63	I	ADC reference voltage : This should be nominally less than or equal to the V _{DD} voltage but should be isolated to minimize noise and error. Level on this pin is used as a reference for ADC(s) and ADC.
V _{BAT}	49	I	RTC power supply voltage : 3.3 V on this pin supplies the power to the RTC.



2.4 System Control Block (PLL and VPB Divider)

2.4.1 PLL

† The PLL accepts an input clock frequency in the range of 10 MHz to 25 MHz. The input frequency is multiplied up into the range of 10 MHz to 60 MHz with a current controlled Oscillator (CCO). The multiplier can be an integer value from 1 to 32 (in practice, the multiplier value cannot be higher than 6 on this family of microcontrollers due to the upper frequency limit of the CPU). The CCO operates in the range of 156 MHz to 320 MHz, so there is an additional divider in the loop to keep the CCO within its frequency range while the PLL is providing the desired output frequency. The output divider may be set to divide by 2, 4, 8 or 16 to produce the output clock. Since the minimum output divider value is 2, it is insured that the PLL output has a 50% duty cycle. The PLL is turned off and bypassed following a chip reset and may be enabled by software. The program must configure and activate the PLL, wait for the PLL to Lock, then connect to the PLL as a clock source. The PLL settling time is 100 μ s.

2.4.2 VPB Divider

† The LPC2141/42/44/46/48 include 2-stage monitoring of the voltage on the V_{DD} pins. If this voltage falls below 2.9 V, the BOD asserts an interrupt signal to the VIC. This signal can be enabled for interrupt; if not, software can monitor the signal by reading dedicated register.

The second stage of low voltage detection asserts reset to inactivate the LPC2141/42/44/46/48 when the voltage on the V_{DD} pins falls below 2.6 V. This reset prevents alteration of the flash as operation of the various elements of the chip would otherwise become unreliable due to low voltage. The BOD circuit maintains this reset down below 1V, at which point the POR circuitry maintains the overall reset.

Both the 2.9V and 2.6V thresholds include some hysteresis. In normal operation, this hysteresis allows the 2.9V detection to reliably interrupt, or a regularly-executed event loop to sense the condition.



2.5 Memory Map

Fig. 2.5.1 shows the different regions in the memory of LPC 2148.

4.0 GB	AHB PERIPHERALS	0xFFFF FFFF
3.75 GB	VPB PERIPHERALS	0xF000 0000
3.5 GB		0xE000 0000
3.0 GB	RESERVED ADDRESS SPACE	0xC000 0000
2.0 GB	BOOT BLOCK (12 kB REMAPPED FROM ON-CHIP FLASH MEMORY)	0x8000 0000
		0x7FFF FFFF
		0x7FFF D000
		0x7FFF CFFF
	RESERVED ADDRESS SPACE	0x7FD0 2000
	8 kB ON-CHIP USB DMA RAM (LPC2148/2148)	0x7FD0 1FFF
		0x7FD0 0000
	RESERVED ADDRESS SPACE	0x7FCF FFFF
		0x4000 8000
	32 kB ON-CHIP STATIC RAM (LPC2148/2148)	0x4000 7FFF
		0x4000 4000
	16 kB ON-CHIP STATIC RAM (LPC2142/2144)	0x4000 3FFF
		0x4000 2000
	8 kB ON-CHIP STATIC RAM (LPC2141)	0x4000 1FFF
1.0 GB		0x4000 0000
	RESERVED ADDRESS SPACE	0x3FFF FFFF
		0x0008 0000
	TOTAL OF 512 kB ON-CHIP NON-VOLATILE MEMORY (LPC2148)	0x0007 FFFF
		0x0004 0000
	TOTAL OF 256 kB ON-CHIP NON-VOLATILE MEMORY (LPC2148)	0x0003 FFFF
		0x0002 0000
	TOTAL OF 128 kB ON-CHIP NON-VOLATILE MEMORY (LPC2144)	0x0001 FFFF
		0x0001 0000
	TOTAL OF 64 kB ON-CHIP NON-VOLATILE MEMORY (LPC2142)	0x0000 FFFF
		0x0000 8000
	TOTAL OF 32 kB ON-CHIP NON-VOLATILE MEMORY (LPC2141)	0x0000 7FFF
0.0 GB		0x0000 0000

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Fig. 2.5.1 : LPC 2141/42/44/46/48 Memory Map († Courtesy NXP)

As seen in the Fig. 2.5.1 locations 0x00000000 to 0x0007FFF is an on-chip non-volatile memory. There after some space is reserved and then there is on-chip static RAM from 0x40000000 to 0x40007FFF. Then there is again some reserved space followed by the USB DMA RAM. This is again followed by reserved space and then the space from 0x7FFFD000 to 0x7FFFFFFF is used by on-chip flash memory.



2.6 † Fast General Purpose Parallel I/O (GPIO)

- Device pins that are not connected to a specific peripheral function are controlled by the GPIO registers. Pins may be dynamically configured as inputs or outputs. Separate registers allow setting or clearing any number of outputs simultaneously. The value of the output register may be read back, as well as the current state of the port pins. LPC2141/42/44/46/48 introduce accelerated GPIO functions over prior LPC2000 devices.
- GPIO registers are relocated to the ARM local bus for the fastest possible I/O timing.
- Mask registers allow treating sets of port bits as a group, leaving other bits unchanged.
- All GPIO registers are byte addressable.
- Entire port value can be written in one instruction.

† Features :

- Bit-level set and clear registers allow a single instruction set or clear of any number of bits in one port.
- Direction control of individual bits.
- Separate control of output set and clear.
- All I/O default to inputs after reset.

2.7 Pin Connect Block

† The pin connect block allows selected pins of the microcontroller to have more than one function. Configuration registers control the multiplexers to allow connection between the pin and the on chip peripherals. Peripherals should be connected to the appropriate pins prior to being activated, and prior to any related interrupt (s) being enabled. Activity of any enabled peripheral function that is not mapped to a related pin should be considered undefined.

The pin control module with its pin select registers defines the functionality of the microcontroller in a given hardware environment.

After reset all pins of Port 0 and Port 1 are configured as input with the following exceptions : If debug is enabled, the JTAG pins will assume their JTAG functionality; if trace is enabled, the Trace pins will assume their trace functionality. The pins associated with the I²C0 and I²C1 interface are open drain.

2.8 Timers

† The Timer/Counter is designed to count cycles of the peripheral clock (PCLK) or an externally supplied clock and optionally generate interrupts or perform other actions at specified timer values, based on four match registers. It also includes four capture inputs to trap the timer value when an input signal transitions. Optionally generating an interrupt. Multiple pins can be selected to perform a single capture or match function, providing an application with 'or' and 'and', as well as 'broadcast' function among them.



The LPC2141/42/44/46/48 can count external events on one of the capture inputs if the minimum external pulse is equal or longer than a period of the PCLK. In this configuration, unused capture lines can be selected as regular timer capture inputs, or used as external interrupts.

Features :

- A 32-bit timer/counter with a programmable 32-bit prescaler.
- External event counter or timer operation.
- Four 32-bit capture channels per timer/counter that can take a snapshot of the timer value when an input signal transitions. A capture event may also optionally generate an interrupt.
- Four 32-bit match registers that allow :
 - Continuous operation with optional interrupt generation on match.
 - Stop timer on match with optional interrupt generation.
 - Reset timer on match with optional interrupt generation.
- Four external output per timer/counter corresponding to match registers, with the following capabilities :
 - Set LOW on match.
 - Set HIGH on match
 - Toggle on match
 - Do nothing on match

2.9 Interfacing Displays to LPC2148

Fast general purpose parallel I/O (GPIO) :

- Device pins that are not connected to a specific peripheral function are controlled by the GPIO registers. Pins may be dynamically configured as inputs or outputs. Separate registers allow setting or clearing any number of outputs simultaneously. The value of the output register may be read back, as well as the current state of the port pins. LPC2141/42/44/46/48 introduce accelerated GPIO functions over prior LPC2000 devices.
- GPIO registers are relocated to the ARM local bus for the fastest possible I/O timing.
- Mask registers allow treating sets of port bits as a group, leaving other bits unchanged.
- All GPIO registers are byte addressable.
- Entire port value can be written in one instruction.

Features :

- Bit-level set and clear registers allow a single instruction set or clear of any number of bits in one port.
- Direction control of individual bits.
- Separate control of output set and clear.
- All I/O default to inputs after reset.

2.9.1 Interfacing Light Emitting Diode (LED) and Seven Segment Display (SSD)

PU - Oct. 2012, May 2015

As shown in the Fig. 2.9.1 the seven segment display uses seven LEDs to make any digit. If all the LEDs are on, it shows the digit 8. There are two types SSDs available.

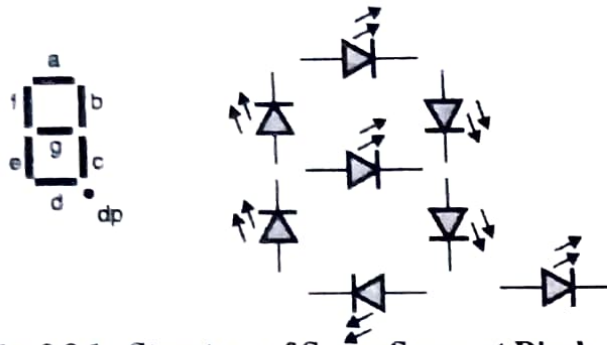


Fig. 2.9.1 : Structure of Seven Segment Display (SSD)

- i) Common cathode i.e. the cathode of all the LEDs are given as a common pin. In this case the anode is connected to the port pins. As already discussed in the section for LEDs, this method requires port pins to source large current. But 8051 cannot source current beyond 2 mA.
- ii) Common anode i.e. the anode of all the LEDs are given as a common pin. In this case the cathode is connected to the port pins. Hence port pins have to sink current of 20 mA.

Hence we use common anode SSD, and connect it to 8051 as shown in Fig. 2.9.2.

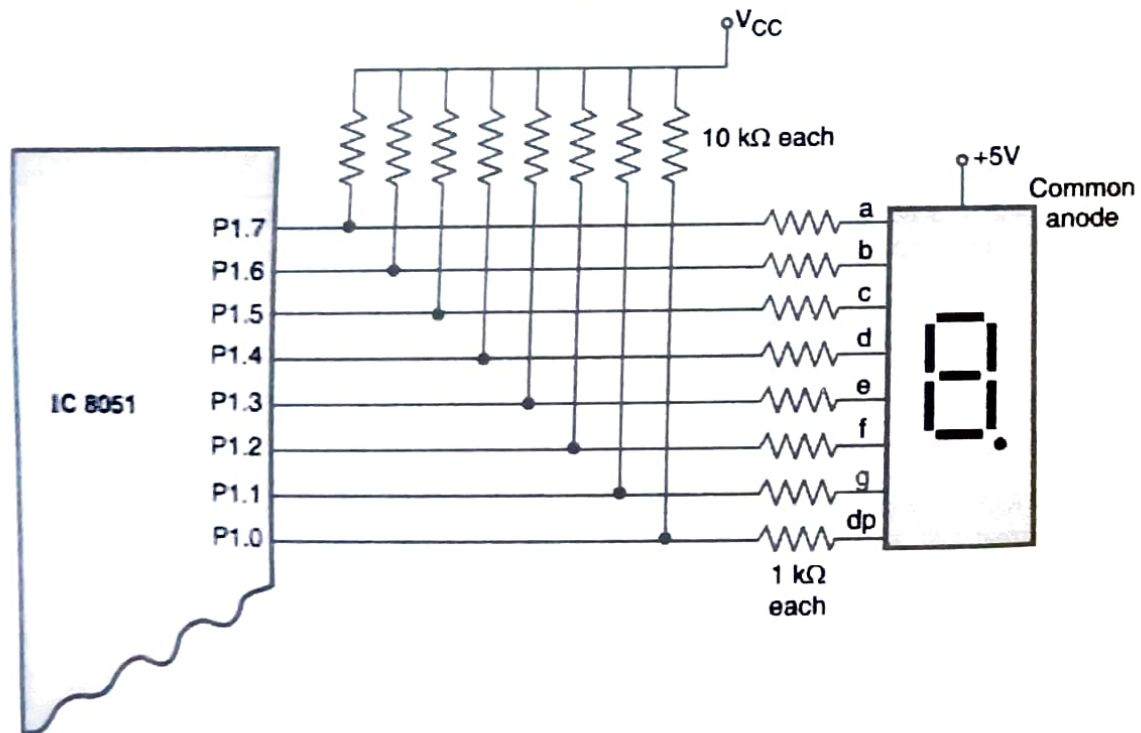
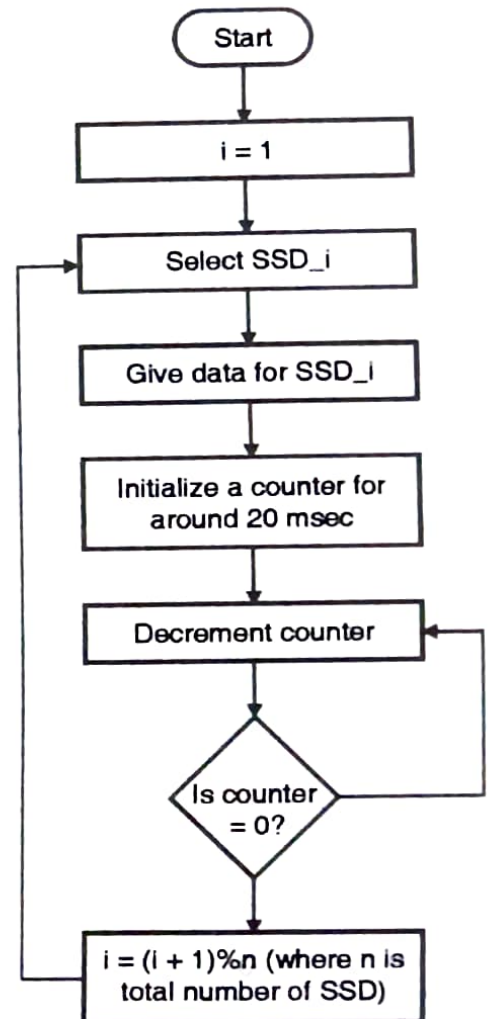


Fig. 2.9.2 : Interfacing common anode SSD to 8051

The 'dp' stands for decimal point. The Table 2.9.1 illustrates the binary and hexadecimal data given to the pins of SSD, to display different digits. A pin should be given logic '0', to switch on the corresponding LED.

Table 2.9.1 : Code for 0 to F given to Common Anode SSD

Digit	g	f	e	d	c	b	a	Hexadecimal form
0	1	0	0	0	0	0	0	0x40
1	1	1	1	1	0	0	1	0x79
2	0	1	0	0	1	0	0	0x24
3	0	1	1	0	0	0	0	0x30
4	0	0	1	1	0	0	1	0x19
5	0	0	1	0	0	1	0	0x12
6	0	0	0	0	0	1	0	0x02
7	1	1	1	1	0	0	0	0x78
8	0	0	0	0	0	0	0	0x00
9	0	0	1	0	0	0	0	0x10
A	0	0	0	1	0	0	0	0x08
B	0	0	0	0	0	1	1	0x03
C	1	0	0	0	1	1	0	0x46
D	0	1	0	0	0	0	1	0x21
E	0	0	0	0	1	1	0	0x06
F	0	0	0	1	1	1	0	0x0E



Flowchart 1

Program 2.9.1 : Write a program to implement two digit counter from 00H to FFH and show the interfacing of the these two SSDs with LPC2148.



Soln. :

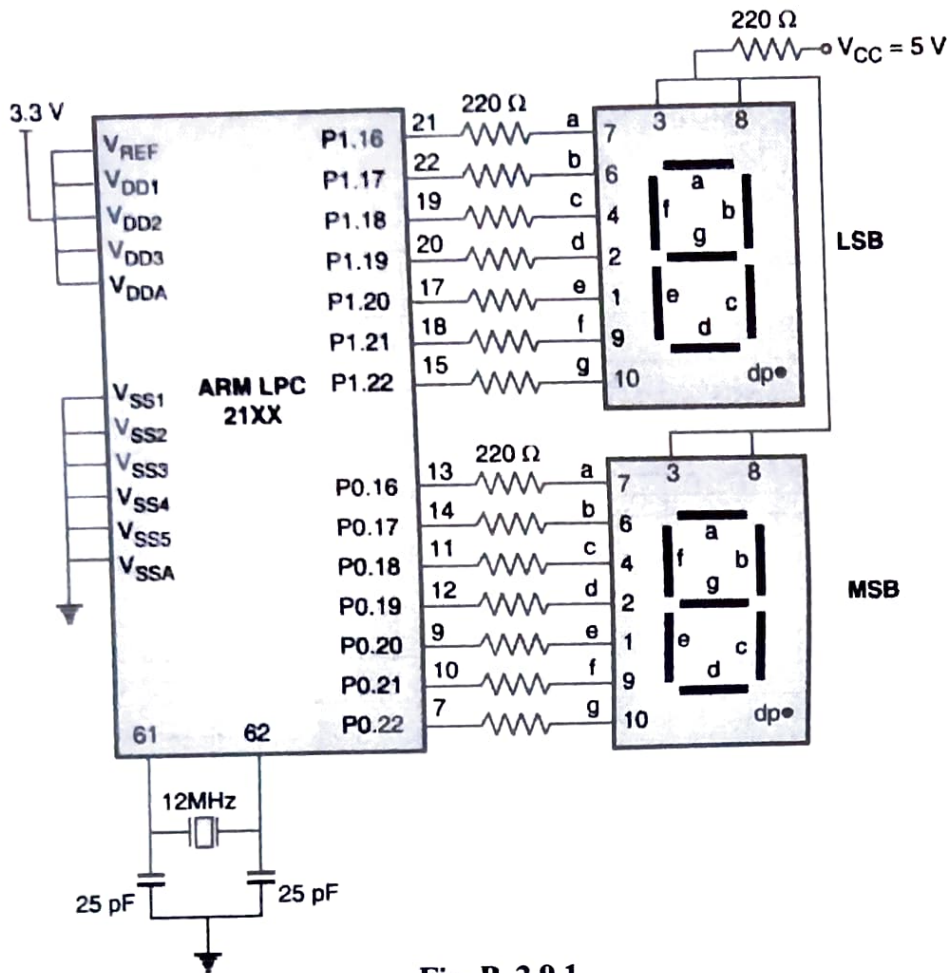


Fig. P. 2.9.1

```
#include <lpc21xx.h>
unsigned char seg[16] = {0x40,0x79,0x24,0x30,0x19,0x12,0x02,0x78,0x00,0x10,0x08,
0x03,0x46,0x21,0x06,0x0E}; //From Table 2.9.1
void main()
{
    unsigned long int k, var=0, var1=0;
    unsigned char seg_val, seg_val1, i, j;
    PINSEL0 = 0X00000000; // Select Port 0 pins as General Purpose I/O lines
    PINSEL1 = 0X00000000; // Select Port 1 pins also as General Purpose I/O lines
    IODIR0 = 0X00FF0000; // Make pins of Port 0 as output pins
    IODIR1 = 0X00FF0000; // Make pins of Port 1 as output pins
    for (i=0; i<=15; i++) // Counter for upper digit
    {
        IOCLR1 = var;
        seg_val = seg[i];
        var = seg_val;
        var = var<<16;
        IOSET1 = var;
        for(j=0; j<=15; j++) // Counter for lower digit
        {
            IOCLR0 = var1;
            seg_val1 = seg[j];
```



```

var1=seg_val1;
var1=var1<<16;
IOSET0=var1;
for(k=0;k<50000;k++);
}
}
}

```

2.9.2 Interfacing LCD

PU - April 2013

- Displaying data in hand help embedded systems should be considering an important aspect, that is the power consumption. LCD displays have very low power consumption compared to the seven segment displays (SSD) and hence are widely used. Another advantage of LCD is that, it uses a 5×7 dot matrix to display, hence better readability is possible. For example to display '1' of LCD as shown in Fig. 2.9.3.
- Also the programming of the LCD is very simple compared to SSD. The message to be displayed on LCD can be very easily given to the LCD controller.

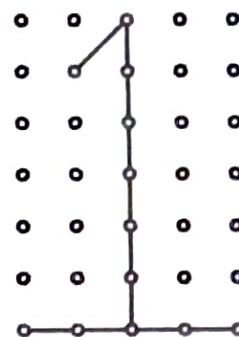


Fig. 2.9.3 : Displaying 1 on LCD display of a 5×7 dot matrix

- The LCDs generally use a common controller chip, Hitachi 44780 and common connector interface. Due to these factors, the alphanumeric LCDs range in size from 8 characters to 80 characters. All the characters are interchangeable without any hardware or software changes. They are arranged in 40 by 2 or 20 by 4 or 10 by 2 or 20 by 1 or 20 by 2. The first figure represents the number of characters in each line and second figure represents the number of lines the display has.

A typical 16 by 2 (i.e. 16 characters and 2 such lines) LCD looks as shown in Fig. 2.9.4.

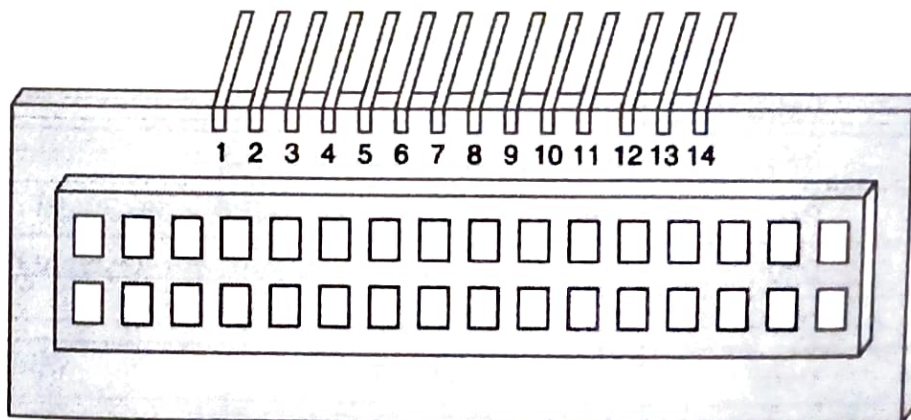


Fig. 2.9.4 : Structures of 16×2 LCD

Some LCD have their pins on left or on bottom. The functions of the pins of LCD are listed in the Table 2.9.2.

Table 2.9.2 : Pin description of LCD

Pins	Symbol	Functions
1	V_{ss}	Ground
2	V_{dd} (or V_{cc})	+ 5 V supply
3	V_o	Contract voltage
4	RS	Should be '0' for instruction and '1' for data.
5	$\overline{R/W}$	Should be '0' to write and '1' to read
6	E	Enable display logic
7	D0	Data bus bit 0
8	D1	Data bus bit 1
9	D2	Data bus bit 2
10	D3	Data bus bit 3
11	D4	Data bus bit 4
12	D5	Data bus bit 5
13	D6	Data bus bit 6
14	D7	Data bus bit 7 (Also used as busy pin)

- There are two registers of LCD viz. Instruction command code register and data register.
- The RS pin is used to select one of these register.
- $\overline{R/W}$ pin is used to read or write to LCD.
- The enable pin E, is used to latch the data into the data or command register. When data is supplied, a high-to-low (negative edge) is required for LCD to latch the data.
- The list of commands that can be given to the LCD are as listed in the Table 2.9.3.

Table 2.9.3 : LCD commands

Hex command	Function
0x01	Clear display
0x02	Return cursor to home
0x04	Decrement cursor (i.e. shift cursor left)
0x06	Increment cursor (i.e. shift cursor right)
0x05	Shift display right
0x07	Shift display left
0x08	Display off, cursor off
0x0A	Display off, cursor on
0x0C	Display on, cursor off

Hex command	Function
0x0E	Display on, cursor on
0x0F	Display on, cursor on and blinking
0x10	Move cursor one position left
0x14	Move cursor one position right
0x18	Shift entire display left
0x1C	Shift entire display right
0x80	Move cursor to beginning of 1 st line
0xC0	Move cursor to beginning of 2 nd line
0x38	Initialize 2 line display of 5 × 7 matrix

PU - April 2013

2.9.3 Initialization of LCD

- The following algorithm is required to initialize and write data to LCD.
- Wait 1 second after power up for display to stabilize.
- Initialize the LCD by giving the instruction 0x38 to the command subroutine.
- Wait for 5 msec.
- Issue the command 0x0F to command subroutine for display on, cursor on and cursor blinking.
- Wait for 5 msec.
- Issue the command 0x01 for clearing display to command subroutine.
- Wait for 5 msec.
- Issue the command 0x06 for making LCD in increment mode i.e. cursor should increment after every character is written to command subroutine.
- Wait for 5 msec.
- Issue the command 0x80 (to command subroutine), to position the cursor at 1st line 1st character.
- Issue the data character one by one giving their ASCII values using data subroutine.

Command subroutine

- Give the instruction to the port connected to data bus of the LCD.
- Make RS = '0', to indicate instruction.
- Make $R/\overline{W}$ = '0', to indicate write.
- Make E = '1'
- Wait for 120 μ sec.
- Make E = '0'
- Return.

} To give a high-to-low pulse on E pin so as to latch the command

**Data subroutine**

- Check if LCD is ready by calling ready subroutine.
- Give the data to the port connected to the data bus of the LCD.
- Make RS = '1', to indicate data
- Make $\overline{R/W}$ = '0', to indicate write
- Make E = '1' } To give a high-to-low
- Wait for 120 μ sec. } pulse on E pin so as
- Make E = '0' } to latch the data
- Return

Ready subroutine

- Make the busy pin (i.e. data bus bit 7) = '1', to program the corresponding port pin of 8051 as input port.
- Make RS = '0' to indicate instruction.
- Make $\overline{R/W}$ = '1', to indicate read.
- Make E = 0
- Make E = 1
- Check if busy pin = '0'. If it is '1', indicates LCD is busy, hence again make E = '0', then E = '1' and check busy pin. Repeat this until busy pin = '0'.
- Return.

Program 2.9.2 : Write a program to display "Embedded Systems" and show the interfacing of LPC2148 with LCD.

Soln. :

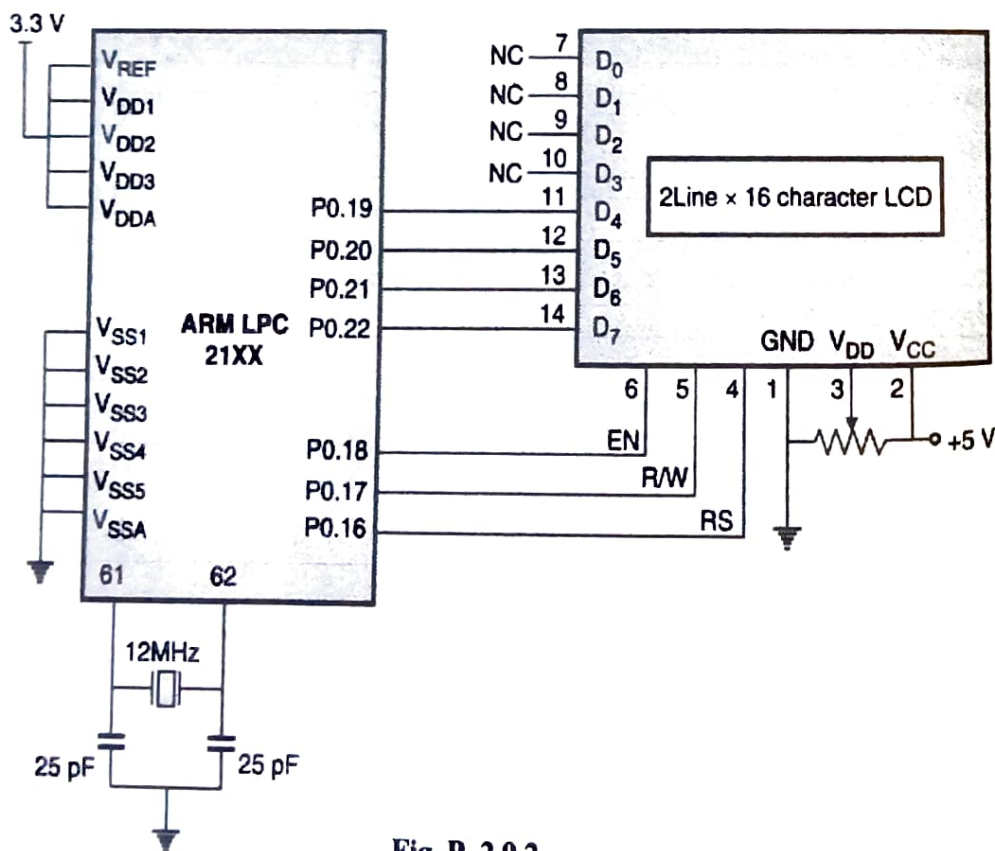


Fig. P. 2.9.2

**Program :**

```
#include <lpc21xx.H>
long unsigned int data,temp1,temp2;
unsigned char *ptr,data_array[20] = "Embedded Systems using ARM";
void lcd_init(void) // Function for initialization of LCD
{
    temp2=0x30; // First command to be given to LCD for initialisation
    temp2=temp2<<16; // Consider lower nibble
    cmd_wrt(); // Call function to write the command
    delay(800); // Call a delay function
    temp2=0x30; // Same command given again for higher nibble
    temp2=temp2<<16;
    cmd_wrt();
    delay(800);
    temp2=0x30;
    temp2=temp2<<16;
    cmd_wrt();
    delay(800);
    temp2=0x30;
    temp2=temp2<<16;
    cmd_wrt();
    delay(800);
    temp2=0x20;
    temp2=temp2<<16;
    cmd_wrt();
    delay(800);
    temp1 = 0x28; // Command for LCD to function in 4 bit mode
    lcd_com();
    delay(800);
    temp1 = 0x0c; // Command for display on, cursor off
    lcd_com();
    delay(800);
    temp1 = 0x06; // Command for cursor increment
    lcd_com();
    delay(500);
    temp1 = 0x80; // Command to force the cursor to beginning of 1st line
    lcd_com();
    delay(800);
}
void delay(unsigned int j) // Delay subroutine
{
    unsigned int k;
    for(k=0;k<j;k++);
}
void clr_disp(void) // Function to clear LCD display
{
    temp1 = 0x01;
```



```
    lcd_com();
    delay(320);
}
void ret_home(void)                                // Function to return home
{
    temp1 = 0x02;
    lcd_com();
    delay(320);
}
void lcd_com(void)                                  // Function for Byte to nibble conversion of LCD
                                                    // command
{
    temp2= temp1 & 0x00f0;
    temp2=temp2<<16;
    cmd_wrt();
    temp2 = temp1 & 0x000f;
    temp2 = temp2 << 20;
    cmd_wrt();
}
void lcd_data(void)                                // Function for Byte to nibble conversion of LCD
                                                    // data
{
    temp2= data & 0x00f0;
    temp2=temp2<<16;
    dat_wrt();
    temp2 = data & 0x000f;
    temp2 = temp2 << 20;
    dat_wrt();
}
void cmd_wrt()
{
    IOSET0=temp2&0x50000;
    delay(500);
    IOSET0=temp2&0x10000;
    delay(500);
}
void dat_wrt()
{
    IOSET0=temp2&0x40000;
    delay(500);
    IOSET0=temp2&0x00000;
    delay(500);
}
void main()
{
    int i=0, size=26;
    PINSEL0 = 0x00000000;
    IODIR0 = 0x00ff0000;
    // Configure Port 0 pins as General Purpose I/O pins
    // Configure Port 0 pins as output pins
```




```

lcd_init();           // Call LCD initialization function
delay(2500);          // Call a delay function
ret_home();           // Call a function for cursor to return home
delay(2500);          // Call a delay function
clr_disp();           // Call a function to clear display
delay(2500);          // Call a delay function
ptr = &data_array[0];
for(i=1;i<size;i++)
{
    if(i == 17)
    {
        temp1 = 0xc0; // Initialize command for shifting cursor to 2nd line
        lcd_com();     // Call the function for converting the byte to nibble
        delay(800);    // Call a delay function
        data = *ptr;   // Call the function for converting the byte to nibble
        lcd_data();     // Point to next character to be displayed
        ptr++;
    }
}
}

```

2.10 Interfacing Keypad

The keyboard is interfaced with microcontroller through input ports. The keyboard consists of mechanical switches. These switches are arranged in non-matrix or matrix form.

2.10.1 Non-matrix Type Keyboard

In non-matrix type keyboard, the key closure is identified by reading the port data, but it requires many port lines. The number of I/O lines is equal to number of keys. Fig. 2.10.1 shows the interfacing of octal non-matrix type keyboard.

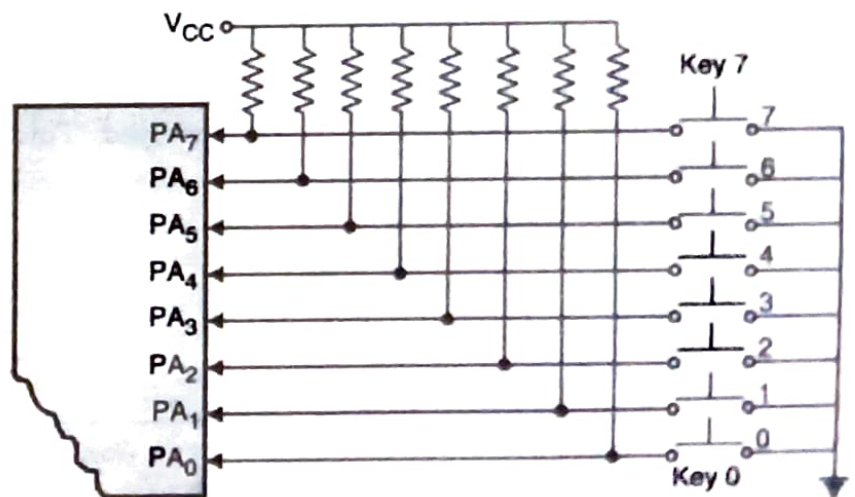


Fig. 2.10.1 : Non matrix type keyboard

To identify the key value the following three functions should be performed :

- (1) Identifying a key closure.
- (2) Debouncing the key.
- (3) Encoding the key to an appropriate code like hexadecimal.



The above three functions can be performed through hardware as well as software. As an example we will see hardware technique for identification of key closure. The interfacing is as shown in Fig. 2.10.2.

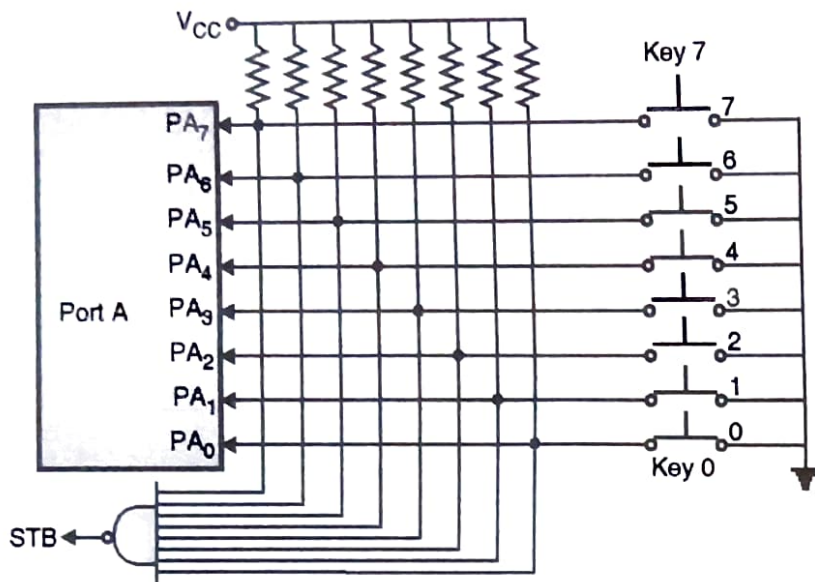


Fig. 2.10.2 : Hardware technique of identification

When all keys are open, the output of NAND gate (STB) goes low. When one of the keys is pressed, the output of NAND gate (STB) goes high. The STB is used to identify that the key is pressed. This STB signal can be used to interrupt the microcontroller.

2.10.2 Matrix Keyboard Interface

PU - April 2012

In simple keyboard, one key is used for interfacing one key each. When the number of keys required are less, it is possible; but when the number of keys required are too many, this method is not possible to be implemented. In such a case different methods like the matrix method of interfacing the keys is used. This reduces the number of port pins required of a microcontroller.

Fig. 2.10.3 shows a 4×4 matrix keyboard i.e. the 16 keys arranged in 4 rows and 4 columns. Fig. 2.10.4 shows the interfacing of a matrix keyboard, it requires two ports : an input port and an output port. The columns are referred to as scan lines and rows are referred to as return lines. If no key is pressed there is no connection between the rows and the columns, but when a key is pressed, there is a connection established between the corresponding row and column and hence a signal is received on the row port pin of the microcontroller, indicating the information of the key pressed.

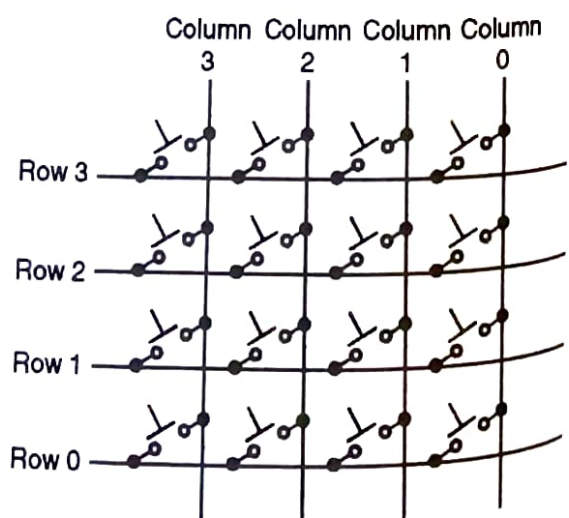


Fig. 2.10.3 : Matrix keyboard

If non matrix type connection is used then 16 lines will be required. So using method reduces the number of connections.

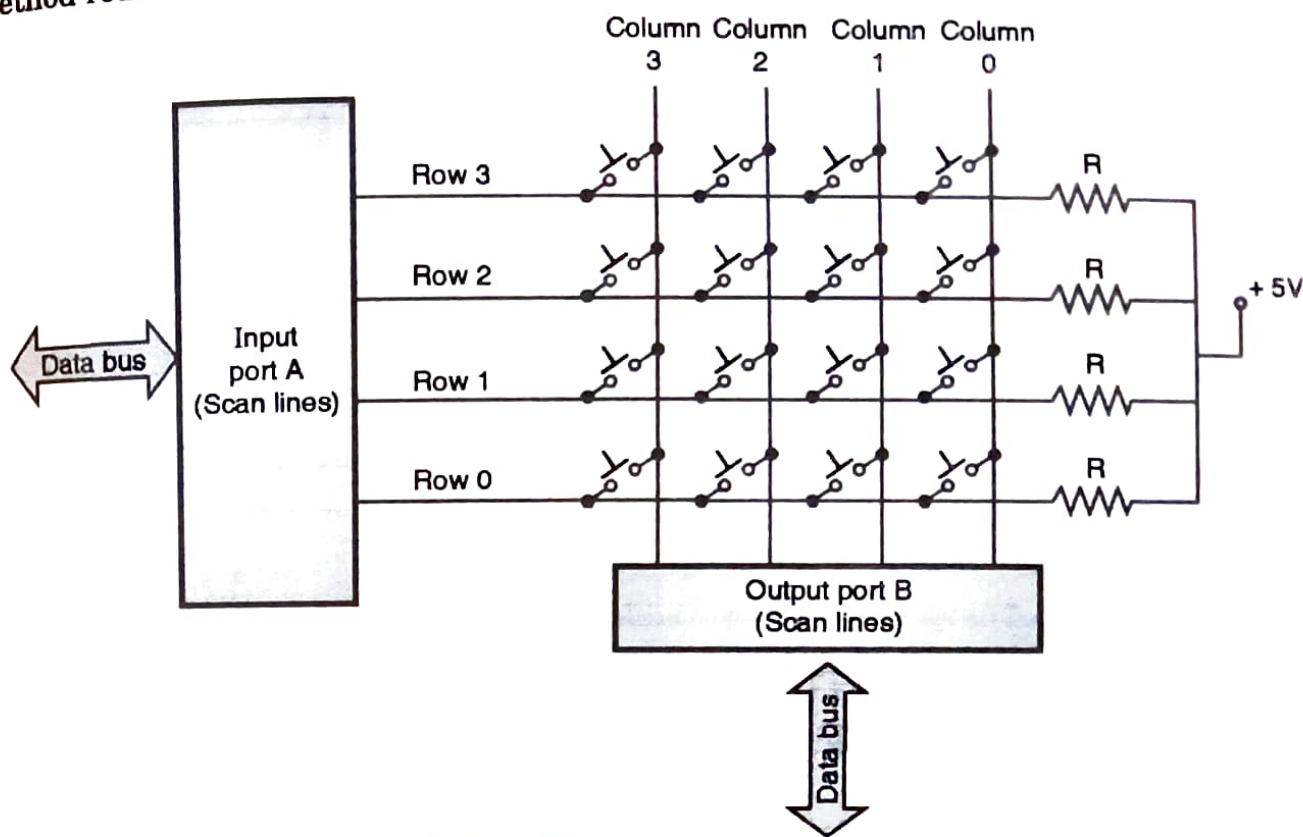


Fig. 2.10.4 : Matrix keyboard connections

The steps required to identify the pressed key are,

- i) To identify if any key is pressed or not.
 - (a) All the column lines are made zero by sending low on all the output lines. i.e. all the keys in the keyboard matrix are activated.
 - (b) Read the status of rows i.e. return lines. If the status of all lines is logic high, the key is not pressed. Otherwise if the status of all lines is logic low, the key is pressed.
- ii) Debouncing the key. (Using software debouncing as explained earlier)
- iii) Identifying the pressed key.
 - (a) Activate the keys from one column by making one column line zero.
 - (b) Read the status of return lines. The zero on any return line indicates that key is pressed.
 - (c) Activate the keys from next column and repeat steps (b) and (c) for all the columns.

2.11 Interfacing GLCD

Most of the LCDs (like JHD12864J) have their displays logically divided into two halves. Hence there are two controllers required with the pin CS1 for controller 1 to drive the left half of the display and the pin CS2 for controller 2 to drive the right half. The two portions for the display are shown in the Fig. 2.11.1. We take the example of a 128 × 64 pixel LCD like JHD12864J.

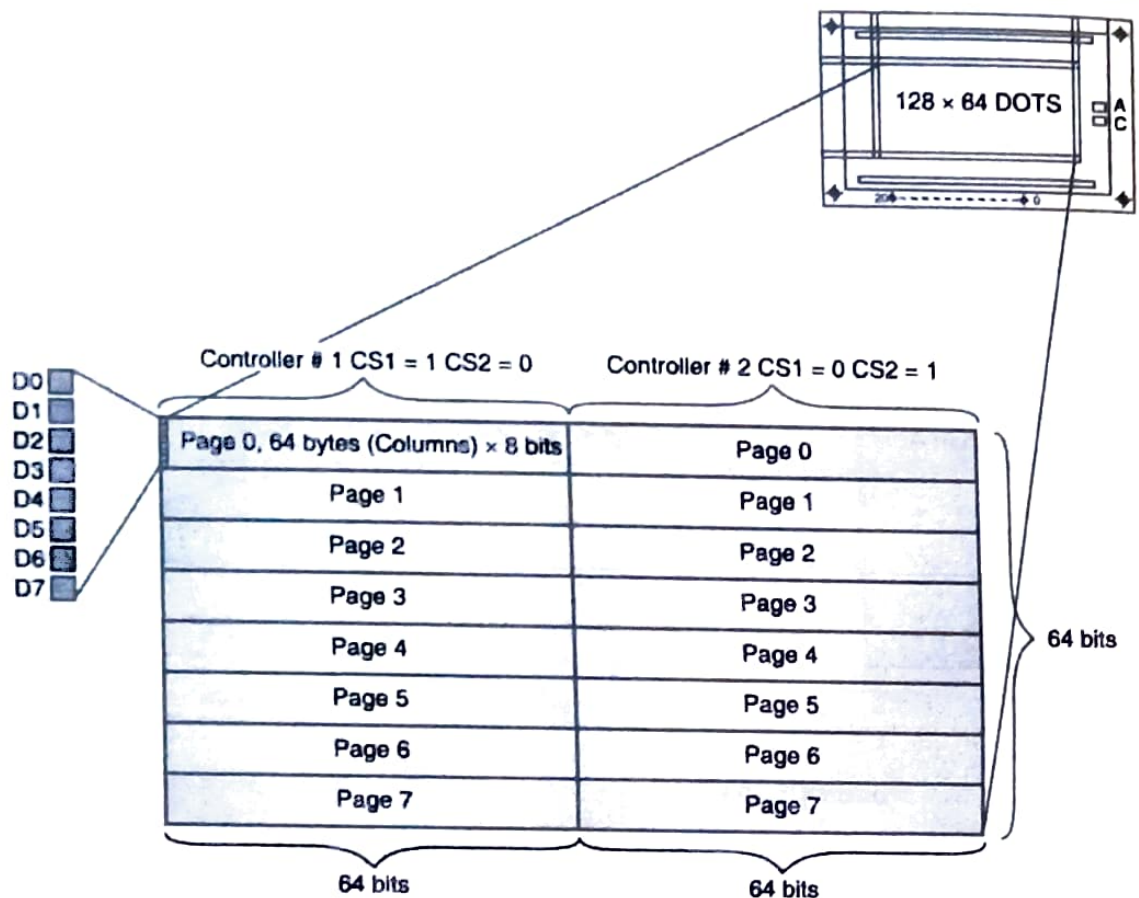


Fig. 2.11.1 : GLCD mapping of the pixels to the controller

The control pins viz. CS1, CS2, R/W and D/I are to be controlled in a proper manner to write the required data in the proper column and row as shown in the Fig. 2.11.1.

There are two column controllers required and one row controller as shown in Fig. 2.11.2. The two controllers KS0108B and KS107B are Samsung controller ICs for the GLCD.

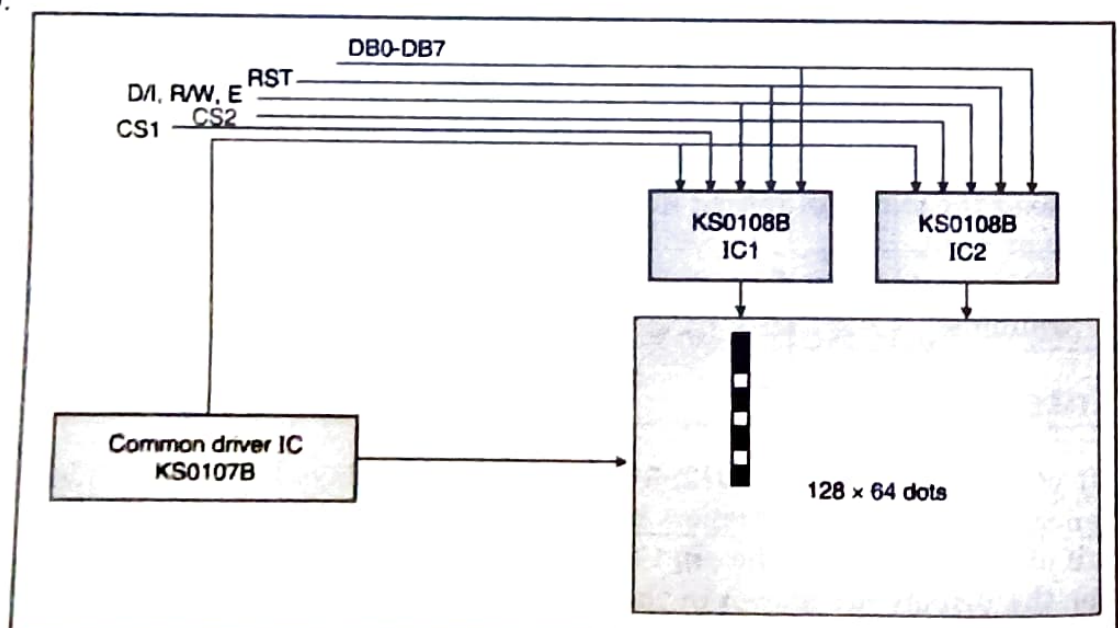


Fig. 2.11.2 : LCD interfacing with the LCD row and column controller



The GLCD can be interfaced with the microcontroller using its port pins. The pins are to be interfaced as shown in the diagram below. The controller select pins CS1 and CS2 are to select the half sections as discussed earlier. The pins D0 - D7, RS, RW and EN have a similar function as that of an LCD (as discussed in section 2.9.2).

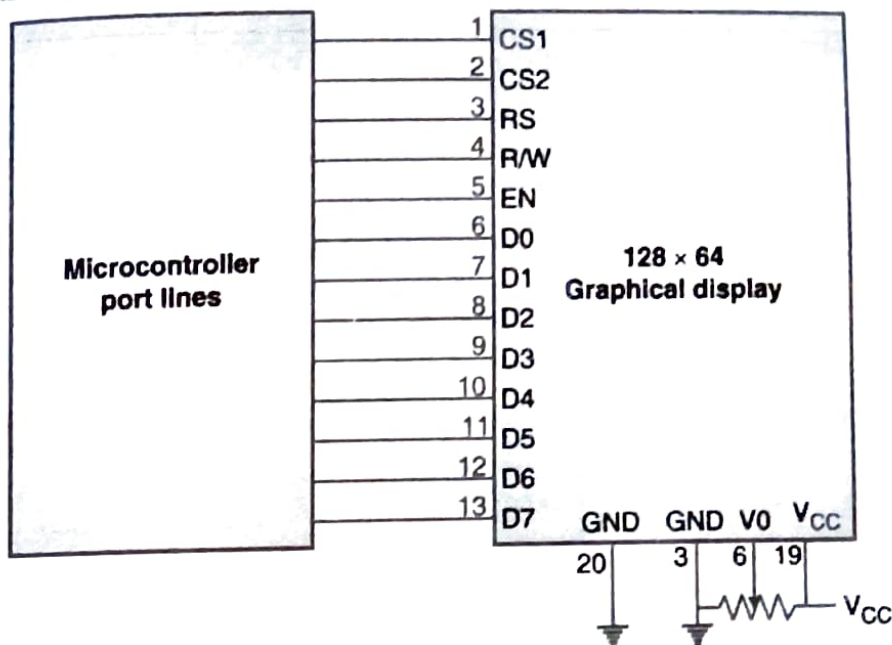


Fig. 2.11.3

2.11.1 Display Control Instruction for KS0108

Instruction	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Function
Display ON/OFF	L	L	L	L	H	H	H	H	H	L/H	Control is display on or off. Internal status end display RAM date is not affected L : OFF, H : ON
Set Address	L	L	L	H	Y address (0-63)						Sets the Y address in the Y address counter
Set Page (X address)	L	L	H	L	H	H	H	Page (0-7)			Sets the X address at the X address register.
Display start line	L	L	H	H	Display start line (0-63)						Indicates the display data RAM displayed at the top of the screen.
Status Read	L	H	B U S Y	L	O N / O F F	R E S E T	L	L	L	L	Read status BUSY L : Ready H : In operation ON/OFF : L : Display ON H : Display OFF RESET L : Normal H : Reset



Instruction	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Function
Write Display data	H	L	Write Data								Writes data (DB0:7) into display data RAM. After writing instruction, Y address is increased by 1 automatically.
Read display data	H	H	Read data								Reads data (DB0:7) from display data RAM to the data bus.

Fig. 2.11.4 : Display control instruction (Samsung KS0108)

2.12 University Questions

Oct. 2011

- Q. 1** Explain with suitable block diagram architecture of LPC2148. (Ans. : Refer section 2.3)

April 2012

- Q. 2** Explain with suitable block diagram architecture of LPC2148.
(Ans. : Refer section 2.3)
- Q. 3** Explain the interfacing of 4×4 Hex keypad to LPC2148 with a diagram.
(Ans. : Refer section 2.10.2)

Oct. 2012

- Q. 4** Explain with suitable block diagram architecture of LPC2148.
(Ans. : Refer section 2.3)
- Q. 5** Draw the interfacing diagram between LPC 2148 and seven segment display.
(Ans. : Refer section 2.9.1)

April 2013

- Q. 6** Draw the interfacing diagram between LPC 2148 and LCD display.
(Ans. : Refer sections 2.9.2 and 2.9.3)

May 2015

- Q. 3(b)** Draw and explain block diagram of LPC 2148. (Ans. : Refer section 2.3) (6 Marks)
- Q. 5(b)** Draw and explain interfacing diagram of 7 segment display with LPC1768. Draw flow chart.
(Ans. : Refer section 2.9.1) (8 Marks)